Iteration #02 Report: Robot Grid Navigation (Outline)
EECE2140 Computing Fundamentals for Engineers, Final Project
Group 9C: Wyatt Zani, Robert, Isaiah Murphy
Repository: https://github.com/WyattZani/Group9c_FinalProject

Status as of Oct 24, 2026 | Due Friday, Oct 30, 2026

Project Leader for Iteration #2 - Robert Krol

1. Project Summary

a. What the program does: user sets grid size (3 to 10), start A, goal B, and obstacle count; obstacles are random but always leave a route; robot uses a 100-unit battery at 2 per move
b. Iteration 1 was one file (Group9FinalDraft1.cpp); this iteration splits it into classes, a menu, and files so each member owns a part
c. Quick status: compiles with zero warnings, every menu option runs; 3 done, 2 in progress, 10 stubs

Robert was Project Leader for this iteration #2
Responsibilities: coordinates daily progress, splits tasks fairly, presents status report at the Wed Oct 28 check-in, keeps task table up to date

3. Task Table

a. Columns: Task ID, name, menu option, responsible, files/functions, deliverable, status
b. Current status:
Menu loop and skeleton: Done
Stub (each prints "not implemented yet")
Exit: Done
README: Done
System design report: Done
Project Plan Excel: In progress

4. System Design

a. Classes
- Point (struct): r, c
- Grid: side, cells; setSize, clear, inBounds, isFree, placeObstacles, pathExists, save, load
- Robot: pos, moves, distance, battery; reset, move, hasBattery, getters
- Navigator: Grid&, Robot, A, B, navigation state; run, step,

getOutcome, getResult
- RunResult (struct): outcome, moves, distance, battery
- AppState (struct): everything the tasks share
- Function modules: display, input, tasks

b. Class diagram (color-coded by owner)
- AppState owns Grid and stores RunResults
- Navigator owns Robot and uses a reference to the same Grid

c. Menu design
- Options 1 to 10 plus 0 (Exit), each calls exactly one task function
- readInt() rejects anything that is not a whole number 0 to 10

d. File structure
- main.cpp, config.h, point.h: Project Leader
- tasks.h/.cpp: Shared (split into sections by member)
- grid.h/.cpp, readPoint() in input.cpp: Isaiah
- robot.h/.cpp, navigator.h/.cpp: Robert
- display.h/.cpp: Wyatt

e. How a run works: options 1 -> 2 -> 3 -> 5 -> 6; result saved to lastRun and history

5. Implementation Progress

a. Skeleton committed; sample console output showing menu and stubs
b. Ownership rule: each member only edits their own section of tasks.cpp and their own files
c. Porting map: which functions from Group9FinalDraft1.cpp move into which file
d. Regular commits to GitHub with clear messages

6. Revised Timeline (Oct 24 to Oct 30)

- Sat 24: Pick leader, agree on task table and class design
- Sat 24: Commit skeleton; README draft
- Tue 27: Options 1 to 3 (Isaiah); options 4 and 7 (Wyatt)
- Tue 27: Option 5 (Robert); option 8 (Wyatt)
- Tue 27: Option 6, start option 10 (Robert); start option 9 (Isaiah); check-in meeting
- Thu 29: Class diagram and design write-up; update Excel; reflections
- Fri 30: Fresh-clone build test; PDF to docs/; final push; each member submits GitHub link on Canvas

7. Team Contributions

a. Wyatt Zani (WyattZani): display, help/legend, reset, README

b. Robert (robk0416): simulation, results, run history
c. Isaiah Murphy (Murphyisa17): grid config, A/B input,
obstacles and route check, save/load
d. Iteration 1 work listed for each member

8. Repository, Build and Run

a. Repo contents: source code, README.md, docs/ folder with PDF and Excel
b. Build: g++ -std=c++17 -Wall src/*.cpp -o robot_navigator
c. Run: ./robot_navigator (Windows: robot_navigator.exe)